

# BTsBM: Bayesian Bradley-Terry Stochastic Block Models in R

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2026-05-22

## Abstract

This software paper documents BTsBM, an R package for Bayesian Bradley-Terry stochastic block models in paired comparison problems. The package is built for settings where strict item-by-item ranking is often too brittle, but an ordered block structure is stable and interpretable. We model directed wins  $w_{ij}$ , infer latent allocations  $x_i$ , infer the number of occupied blocks  $K$ , and estimate block intensities  $\lambda_k$ . The package provides Gibbs samplers, post-processing for label alignment, visual diagnostics, and model comparison tools based on leave-one-out criteria.

The paper has two aims. The first is methodological: to explain the model and the Markov chain Monte Carlo updates in the notation used across the package. The second is practical: to provide a reproducible workflow with code and output, in the style commonly used in R package vignettes.

```
library(BTsBM)
```

## Introduction

This model is for paired-comparison analyses where the data are directional outcomes between items, such as wins/losses in sport, preference judgements, or expert comparisons. The package expects these data in adjacency-matrix form: a directed count matrix  $\mathbf{W} = (w_{ij})$ , where  $w_{ij}$  is how many times  $i$  is preferred to or beats  $j$ .

The standard Bradley-Terry model gives a strict item-level ranking. In many applications that resolution is too fine: nearby items are often not credibly separable. BT-SBM keeps the Bradley-Terry comparison structure but ranks latent blocks rather than individuals, so items in the same block are rank-indifferent while blocks remain ordered by strength.

## Quick start

```
library(BTsBM)

# Directed win-count matrix (n x n), with zero diagonal
w_ij <- ATP_2000_2025$`2017`$Y_ij
diag(w_ij) <- 0L

fit <- gibbs_bt_sbm(
  w_ij = w_ij,
  prior = "GN",
  gamma_GN = 0.8,
  T_iter = 400,
  T_burn = 200,
  verbose = FALSE
```

)

```
post <- relabel_by_lambda(fit$x_samples, fit$lambda_samples)
plot_block_adjacency(fit = post, w_ij = w_ij)
```

## Data formatting details (optional)

When raw data come with multiple contexts or judges, aggregate them before fitting to produce  $\mathbf{W}$ . Let  $c \in \{1, \dots, C\}$  index context,  $j \in \{1, \dots, J\}$  index judge/rater, and  $r$  index repeated pairwise observations. Define

$$y_{ij}^{(c,j,r)} = \mathbf{1}\{\text{in replicate } r, \text{ judge } j \text{ in context } c \text{ prefers } i \text{ to } j\}.$$

Then

$$w_{ij}^{(c,j)} = \sum_r y_{ij}^{(c,j,r)},$$

and the package input is

$$w_{ij} = \sum_{c=1}^C \sum_{j=1}^J w_{ij}^{(c,j)}, \quad n_{ij} = w_{ij} + w_{ji}.$$

This gives the directed adjacency matrix  $\mathbf{W}$  used by `gibbs_bt_sbm()`. If context effects are central, repeat this construction within each context  $c$  and fit context-specific models.

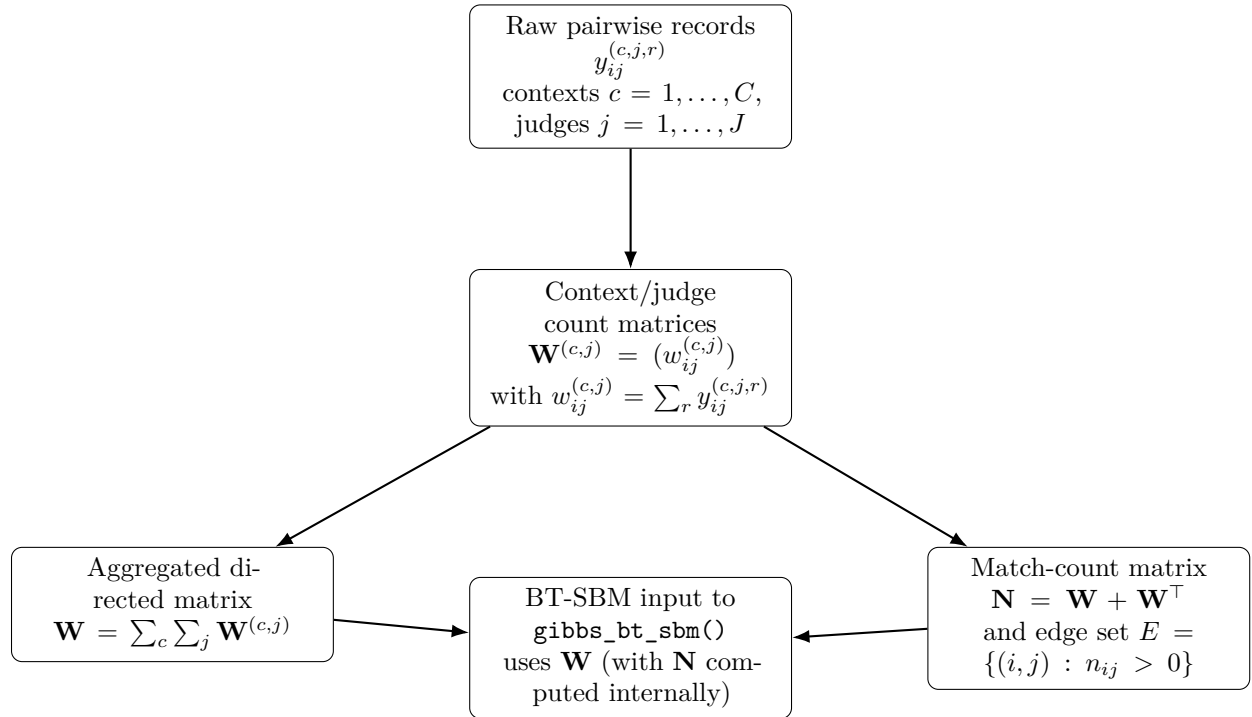


Figure 1: Data-construction map for context/judge pairwise data. Raw comparisons are first summarized into context- and judge-specific count matrices, then aggregated into the adjacency matrix  $\mathbf{W}$  used by BT-SBM.

## Model and Notation

We use the same notation as the main BT-SBM paper and keep a one-to-one map to package objects.

Let  $w_{ij}$  be directed wins, with  $w_{ii} = 0$ , and define

$$n_{ij} = w_{ij} + w_{ji}.$$

Collecting all  $w_{ij}$  gives the directed outcome matrix  $\mathbf{W} \in \mathbb{N}^{n \times n}$ , while  $\mathbf{N} = \mathbf{W} + \mathbf{W}^\top$  is symmetric with zero diagonal. We also use  $E = \{(i, j) : n_{ij} > 0\}$  for observed pairs.

Each item  $i$  has a latent block label  $x_i \in \{1, \dots, K\}$ . Each occupied block  $k$  has intensity  $\lambda_k > 0$ . The Bradley-Terry map is

$$\theta_{x_i x_j} = \frac{\lambda_{x_i}}{\lambda_{x_i} + \lambda_{x_j}},$$

and outcomes satisfy

$$w_{ij} \mid n_{ij}, x, \lambda \sim \text{Binomial}(n_{ij}, \theta_{x_i x_j}).$$

For priors on  $\lambda_k$ , BTsBM uses

$$\lambda_k \sim \text{Gamma}(a, b), \quad b = \exp\{\psi(a)\},$$

so that  $\mathbb{E}[\log \lambda_k] = 0$  a priori.

For partitions, the package supports DP, PY, DM, and GN urn-style priors through prior weights in the single-site allocation updates.

To make the translation to code explicit, the paper notation  $(\mathbf{W}, \mathbf{N}, x, \lambda, K, E)$  corresponds to `(w_ij, w_ij + t(w_ij), x_samples, lambda_samples, K_per_iter, obs_idx)` in package outputs.

## Priors on $x$ : what is supported and how to set it

The prior on allocations  $x = (x_1, \dots, x_n)$  is selected with the `prior` argument in `gibbs_bt_sbm()`. The supported options are:

| prior | Hyperparameters                 | Internal helper     | Typical use                                           |
|-------|---------------------------------|---------------------|-------------------------------------------------------|
| "DP"  | <code>alpha_PY</code>           | <code>urn_DP</code> | Baseline CRP-like reinforcement                       |
| "PY"  | <code>alpha_PY, sigma_PY</code> | <code>urn_PY</code> | Heavier-tail clustering than DP                       |
| "DM"  | <code>beta_DM, K_DM</code>      | <code>urn_DM</code> | Finite-cap clustering with max <code>K_DM</code>      |
| "GN"  | <code>gamma_GN</code>           | <code>urn_GN</code> | Finite-species Gibbs-type prior (default in examples) |

Below are the corresponding predictive weights (as in Legramanti, Rigon, Durante, and Dunson, 2022):

1. `prior = "DP"` (Dirichlet process / CRP-type weights) Uses `alpha_PY` and ignores `sigma_PY`:

$$(v_1, \dots, v_K, \alpha_{PY}),$$

implemented by `urn_DP(v_minus, alpha_PY)`.

2. `prior = "PY"` (Pitman-Yor) Uses `alpha_PY` and `sigma_PY` \in (0,1):

$$(v_1 - \sigma_{PY}, \dots, v_K - \sigma_{PY}, \alpha_{PY} + K\sigma_{PY}),$$

implemented by `urn_PY(v_minus, alpha_PY, sigma_PY)`.

3. `prior = "DM"` (finite Dirichlet-Multinomial) Uses `beta_DM` and a finite cap `K_DM`:

$$(v_1 + \beta_{DM}, \dots, v_K + \beta_{DM}, \beta_{DM}(K_{DM} - K)\mathbf{1}_{K < K_{DM}}),$$

implemented by `urn_DM(v_minus, beta_DM, K_DM)`.

4. `prior = "GN"` (Gnedin-type finite-species prior) Uses `gamma_GN`:

$$((v_1 + 1)(n - K + \gamma), \dots, (v_K + 1)(n - K + \gamma), K^2 - K\gamma),$$

implemented by `urn_GN(v_minus, gamma_GN)`.

In code, `gibbs_bt_sbm()` validates the needed hyperparameters and then dispatches through `urn_fun <- switch(prior, DP = ..., PY = ..., DM = ..., GN = ...)`. During each single-site update of  $x_i$ , this gives the prior mass for assigning item  $i$  to each existing block and to one candidate new block.

Minimal usage examples:

```
# Dirichlet process prior on x
fit_dp <- gibbs_bt_sbm(w_ij, prior = "DP", alpha_PY = 1.0)

# Pitman-Yor prior on x
fit_py <- gibbs_bt_sbm(w_ij, prior = "PY", alpha_PY = 1.0, sigma_PY = 0.3)

# Finite Dirichlet-Multinomial prior on x (at most K_DM blocks)
fit_dm <- gibbs_bt_sbm(w_ij, prior = "DM", beta_DM = 0.5, K_DM = 15)

# Gnedin prior on x
fit_gn <- gibbs_bt_sbm(w_ij, prior = "GN", gamma_GN = 0.8)
```

These four choices correspond directly to the helpers `urn_DP`, `urn_PY`, `urn_DM`, and `urn_GN` used internally by the sampler, and are the package-level interface to the partition-prior family discussed in Legramanti, Rigon, Durante, and Dunson (2022).

## Posterior Computation

The core sampler in `gibbs_bt_sbm()` alternates three main blocks.

First, latent variables  $Z_{ij}$  are sampled for observed pairs  $(i, j)$  with  $n_{ij} > 0$ :

$$Z_{ij} \sim \text{Gamma}\left(n_{ij}, \frac{1}{\lambda_{x_i} + \lambda_{x_j}}\right),$$

with symmetry  $Z_{ij} = Z_{ji}$ . This step is implemented in C++ for speed.

Second, each label  $x_i$  is updated by single-site Gibbs moves. Existing clusters and one new-cluster option are compared on a log-weight scale:

$$\log p_h \propto \log(\text{prior mass}_h) + \text{likelihood contribution}_h.$$

For existing clusters, the key term is

$$w_i \log \lambda_h - \lambda_h \sum_j Z_{ij}, \quad w_i = \sum_j w_{ij}.$$

For the new cluster, the package uses the integrated score under the Gamma prior.

Third, occupied  $\lambda_k$  are updated conjugately:

$$\lambda_k | - \sim \text{Gamma}\left(a + \sum_{i:x_i=k} w_i, b + \sum_{i:x_i=k} \sum_j Z_{ij}\right).$$

Then a global geometric-mean rescaling of occupied intensities is applied for numerical stability.

## Software Design

The implementation separates fitting, utilities, and plotting.

- `gibbs_bt_sbm()` is the clustered sampler.
- `gibbs_bt_simple()` is the item-level Bradley-Terry baseline.
- `relabel_by_lambda()` aligns labels across MCMC draws and computes posterior partition summaries.
- `plot_block_adjacency()`, `plot_assignment_probabilities()`, `plot_lambda_uncertainty()`, and `plot_rank_intervals()` provide structured diagnostics.
- `make_bt_simple_loo()`, `make_bt_cluster_loo()`, and `compare_bt_models_loo()` provide out-of-sample comparison helpers.

The package ships with `ATP_2000_2025`, a yearly list with  $Y_{ij}$ ,  $N_{ij}$ , and player metadata.

## Reproducible Workflow

### Simulation-first example

To ensure clear multi-cluster structure in a short run, we begin with synthetic paired-comparison data generated from three latent blocks.

```
sim <- sample_from_BTSM(
  n_players = 18,
  K = 3,
  mean_matches = 8,
  p_edge = 0.75,
  lambda = c(0.25, 0.9, 2.4),
  reverse_lambda = FALSE,
  seed = 123
)

w_ij <- sim$w
diag(w_ij) <- 0L
player_ids <- paste0("Player_", seq_len(nrow(w_ij)))
dimnames(w_ij) <- list(player_ids, player_ids)

w_ij[1:6, 1:6]
#>      Player_1 Player_2 Player_3 Player_4 Player_5 Player_6
#> Player_1      0       1       0       0       0       0
#> Player_2      5       0       1       6       0       0
#> Player_3      0       6       0       0       5       2
#> Player_4      5       2       0       0       0       0
#> Player_5      0       0       4      12       0       0
#> Player_6      0       0       5       0       5       0

n <- nrow(w_ij)
S_demo <- 300 - 150
c(n = n, S = S_demo)
#>   n   S
#> 18 150
```

### Fit clustered model

```
fit <- gibbs_bt_sbm(
  w_ij = w_ij,
```

```

a = 2,
prior = "GN",
gamma_GN = 0.8,
T_iter = 300,
T_burn = 150,
verbose = FALSE
)

str(list(
  x_samples = dim(fit$x_samples),
  K_per_iter = length(fit$K_per_iter),
  lambda_samples = length(fit$lambda_samples)
))
#> List of 3
#> $ x_samples      : int [1:2] 150 18
#> $ K_per_iter     : int 150
#> $ lambda_samples : int 150

head(sort(table(fit$K_per_iter), decreasing = TRUE), 6)
#>
#>  3  4  5  6  7 10
#> 94 32 14  7  1  1

```

## Relabel and summarize

```

post <- relabel_by_lambda(fit$x_samples, fit$lambda_samples)
if (knitr::is_latex_output()) {
  knitr::kable(
    post$block_count_distribution,
    caption = "Posterior distribution of the number of occupied blocks"
  )
} else {
  pretty_table_K_distribution(post)
}

```

Table 2: Posterior distribution of the number of occupied blocks

| num_blocks | count | prob      |
|------------|-------|-----------|
| 3          | 94    | 0.6266667 |
| 4          | 32    | 0.2133333 |
| 5          | 14    | 0.0933333 |
| 6          | 7     | 0.0466667 |
| 7          | 1     | 0.0066667 |
| 10         | 1     | 0.0066667 |
| 11         | 1     | 0.0066667 |

```

head(post$block_count_distribution)
#>   num_blocks count      prob
#> 1          3    94 0.6266667
#> 2          4    32 0.2133333
#> 3          5    14 0.0933333
#> 4          6     7 0.0466667

```

```
#> 5          7      1 0.006666667
#> 6         10      1 0.006666667

length(unique(post$minVI_partition))
#> [1] 3
```

## Visual diagnostics

```
plot_block_adjacency(fit = post, w_ij = w_ij)
```

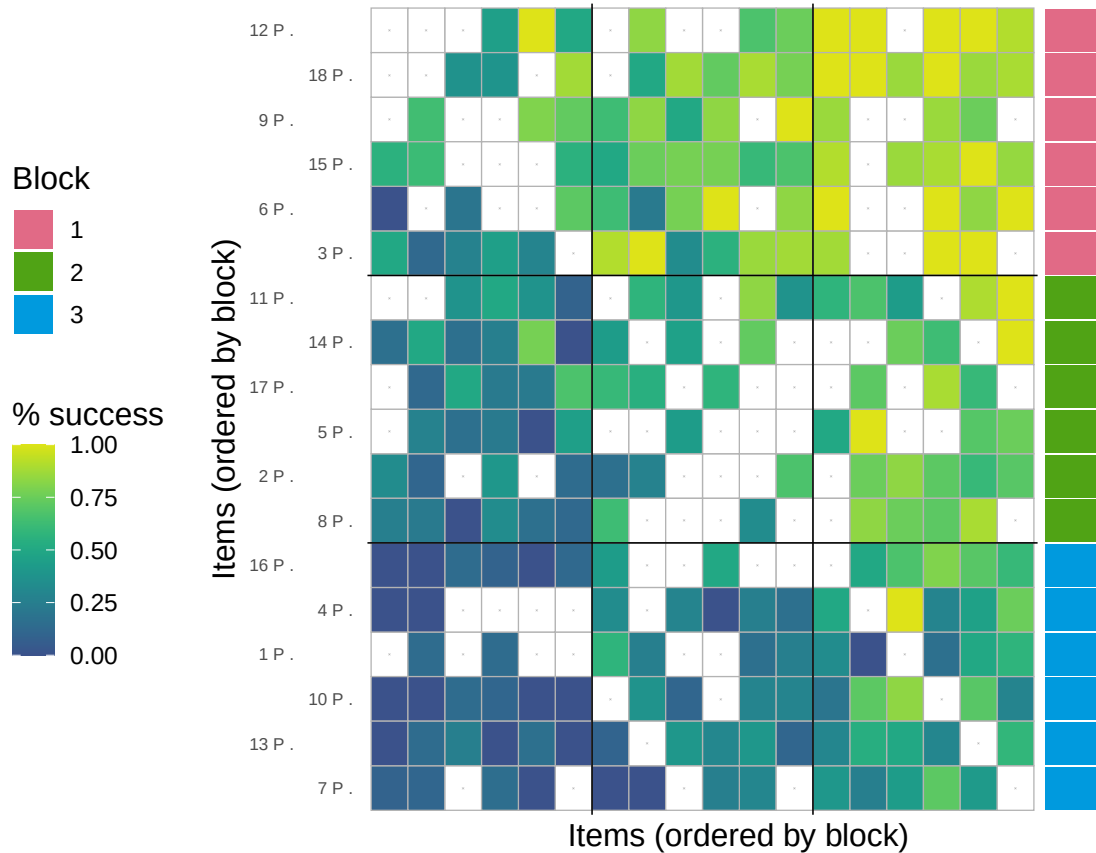


Figure 2: Block-ordered adjacency heatmap. Rows and columns are ordered by inferred blocks, with boundary lines indicating block transitions.

```
plot_assignment_probabilities(fit = post, w_ij = w_ij, x_hat = post$minVI_partition)
plot_lambda_uncertainty(fit = post, w_ij = w_ij, x_hat = post$minVI_partition)
plot_rank_intervals(post, max_players = 10)
```

## Small ATP illustration

The simulation above is used for cluster-recovery diagnostics. For a lightweight real-data illustration, we run the same workflow on a small ATP subset.

```
w_ij_atp <- ATP_2000_2025$`2017`$Y_ij[1:10, 1:10]
diag(w_ij_atp) <- 0L
```

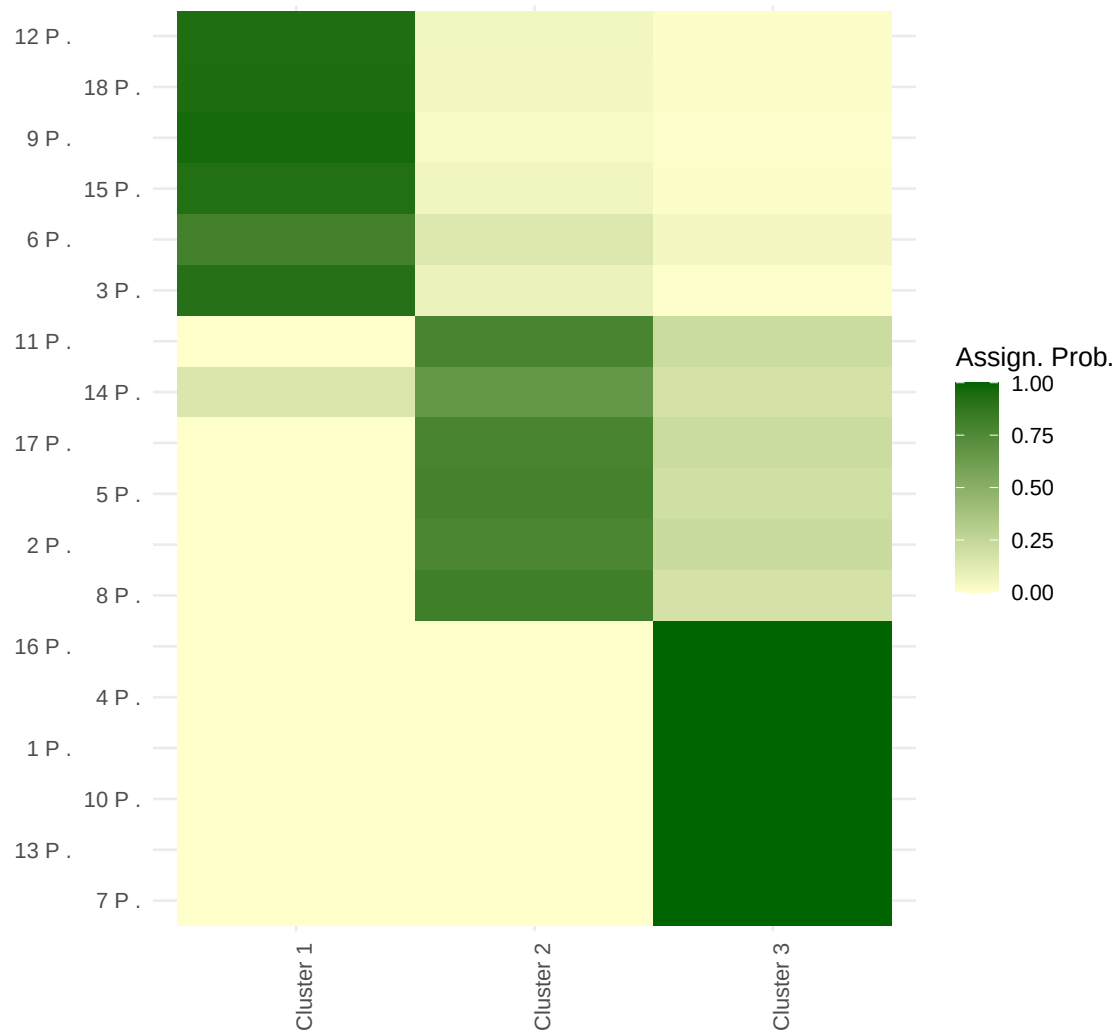


Figure 3: Posterior assignment probabilities by player and cluster, after relabelling for interpretability.



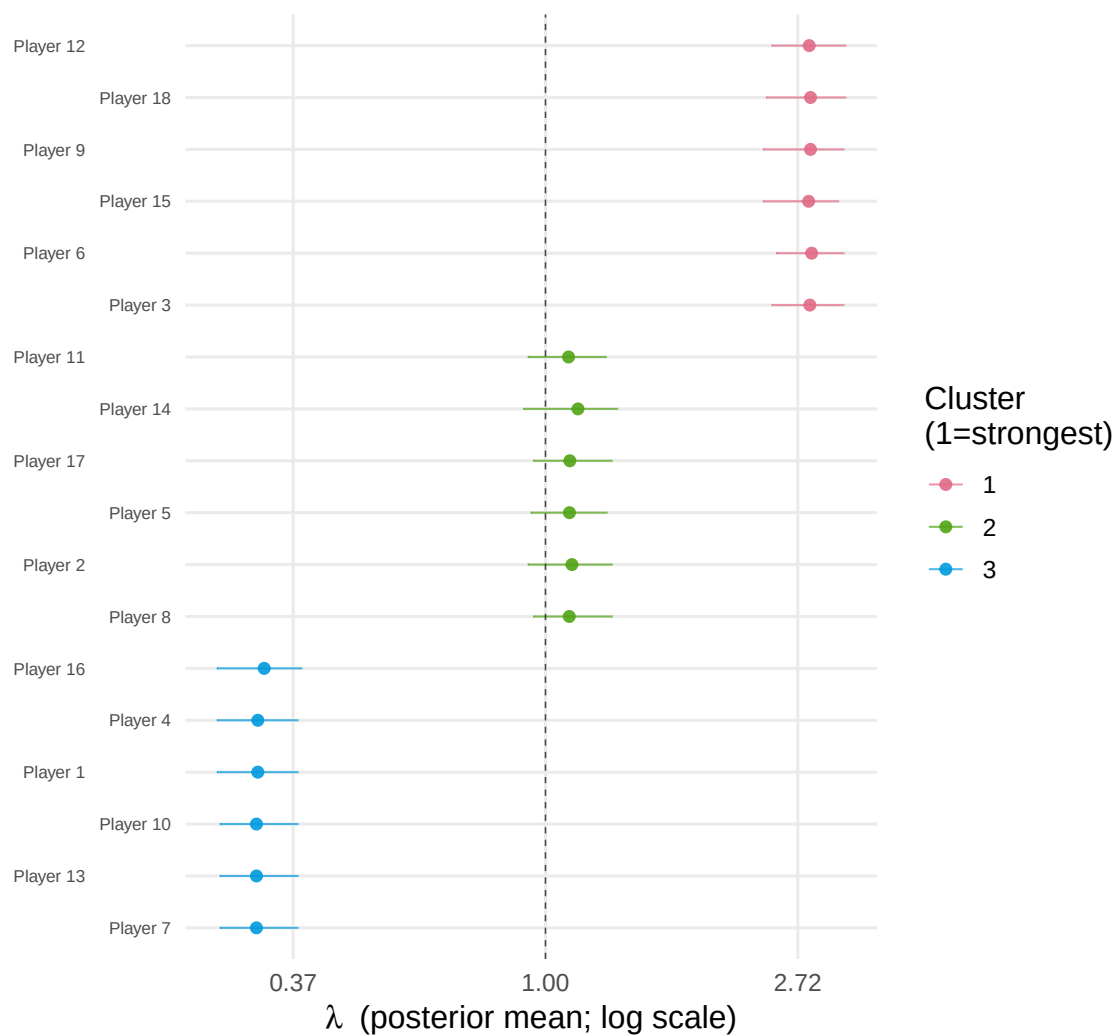


Figure 4: Posterior uncertainty in block intensity parameters (lambda), summarising relative block strengths.

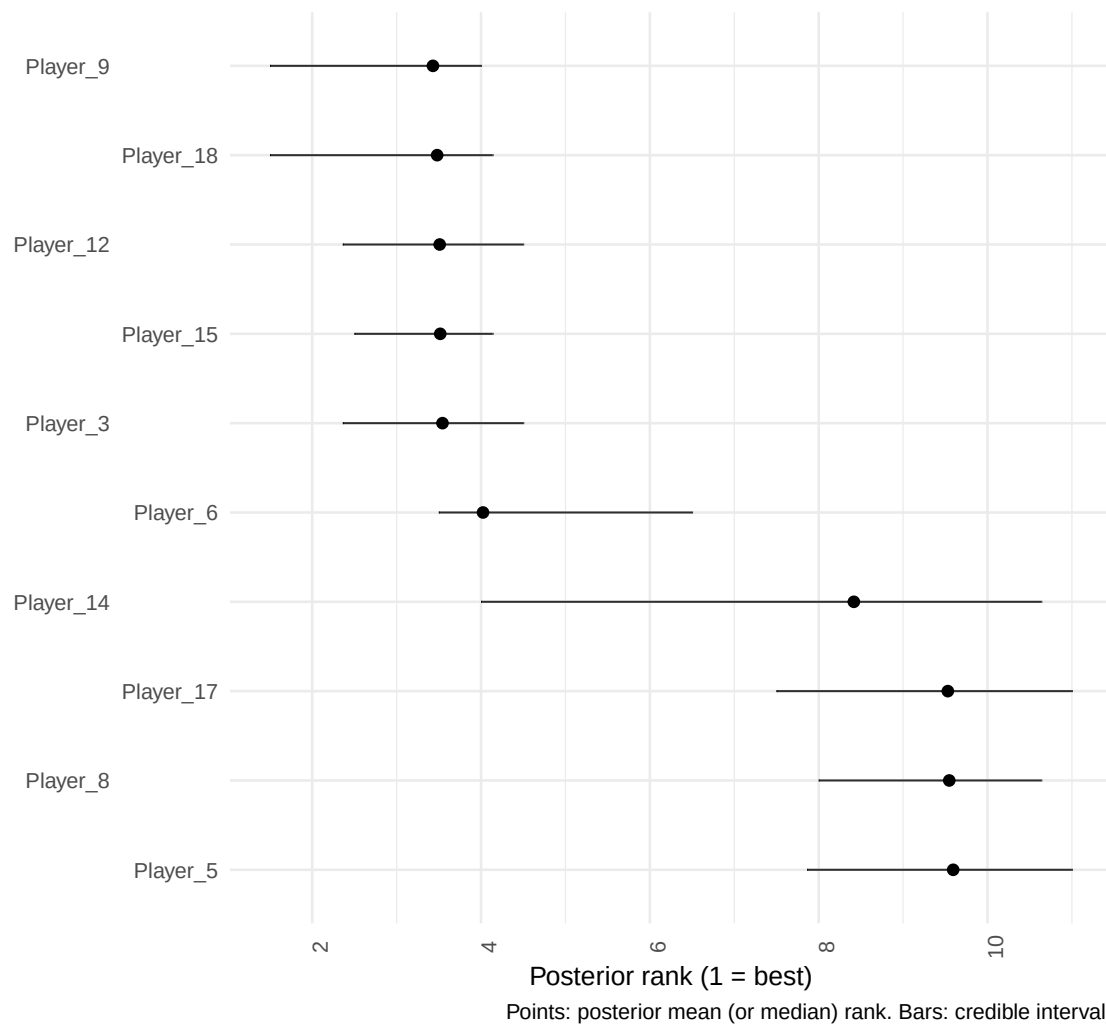


Figure 5: Posterior rank intervals for the top players, highlighting uncertainty in the induced ranking.

```

atp_ids <- rownames(w_ij_atp)
if (is.null(atp_ids)) {
  atp_ids <- paste0("ATP_", seq_len(nrow(w_ij_atp)))
}
dimnames(w_ij_atp) <- list(atp_ids, atp_ids)

fit_atp <- gibbs_bt_sbm(
  w_ij = w_ij_atp,
  a = 2,
  prior = "GN",
  gamma_GN = 0.8,
  T_iter = 180,
  T_burn = 90,
  verbose = FALSE
)

post_atp <- relabel_by_lambda(fit_atp$x_samples, fit_atp$lambda_samples)
head(post_atp$block_count_distribution)
#>   num_blocks count      prob
#> 1           2    76 0.84444444
#> 2           3    10 0.11111111
#> 3           4     3 0.03333333
#> 4           5     1 0.01111111

```

## Predictive comparison with simple BT

```

fit_simple <- gibbs_bt_simple(
  w_ij = w_ij,
  a = 0.01,
  b = 0.1,
  T_iter = 300,
  T_burn = 150,
  verbose = FALSE
)

llo_simple <- make_bt_simple_loo(w_ij, fit_simple$lambda_samples)
llo_cluster <- make_bt_cluster_loo(
  w_ij,
  post$lambda_samples_relabel,
  post$x_samples_relabel
)

# The compare step requires the suggested package 'loo'.
if (requireNamespace("loo", quietly = TRUE)) {
  cmp_out <- tryCatch(
    compare_bt_models_loo(llo_simple, llo_cluster),
    error = function(e) e
  )

  if (inherits(cmp_out, "error")) {
    paste(
      "LOO comparison is unavailable in this local setup:",
      conditionMessage(cmp_out)
    )
  }
}

```

```

    )
  } else {
    cmp_out$comparison
  }
} else {
  "Install 'loo' to run compare_bt_models_loo()."
}
#>           elpd_diff se_diff
#> model2         0.0      0.0
#> model1 -1359.9    382.6

```

## Practical Notes for Package Use

The object dimensions used by BTSBM are consistent across functions. For  $S = T_{\text{iter}} - T_{\text{burn}}$ , `x_samples` has shape  $S \times n$ , `K_per_iter` has length  $S$ , and relabelled item-level intensities are stored as `lambda_samples_relabel` with shape  $S \times n$ .

The main plotting functions expect relabelled output, not raw labels, because raw cluster indices are permutation-invariant across iterations.

For computational control, the most important switches are matrix size  $n$ , number of saved draws  $S$ , and whether latent  $Z$  draws are stored. The `store_z = TRUE` option is useful for diagnostics but substantially increases memory use.

## Assumptions and Limitations

BTSBM is intentionally focused. Outcomes are modelled through Bradley-Terry probabilities determined by latent intensities; pair-specific covariates and dynamic effects are not in the current core model. Partition priors are exchangeable and encoded through urn-style updates. The approach is effective for moderate problem sizes common in sports and ranking applications, but very large dense comparison matrices may require additional engineering.

## Conclusion

BTSBM provides a coherent workflow for Bayesian block-structured ranking from paired comparisons. It combines a model with interpretable latent structure, practical posterior computation, and publication-oriented diagnostics. The package is especially useful when strict total rankings are unstable but tiered structure is statistically and substantively credible.

## References

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- Ferguson, T. S. (1973). A Bayesian analysis of some nonparametric problems. *The Annals of Statistics*, 1(2), 209-230.
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- Legramanti, S., Rigon, T., Durante, D., and Dunson, D. B. (2022). Extended stochastic block models with application to criminal networks. *The Annals of Applied Statistics*, 16(2), 865-891.

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